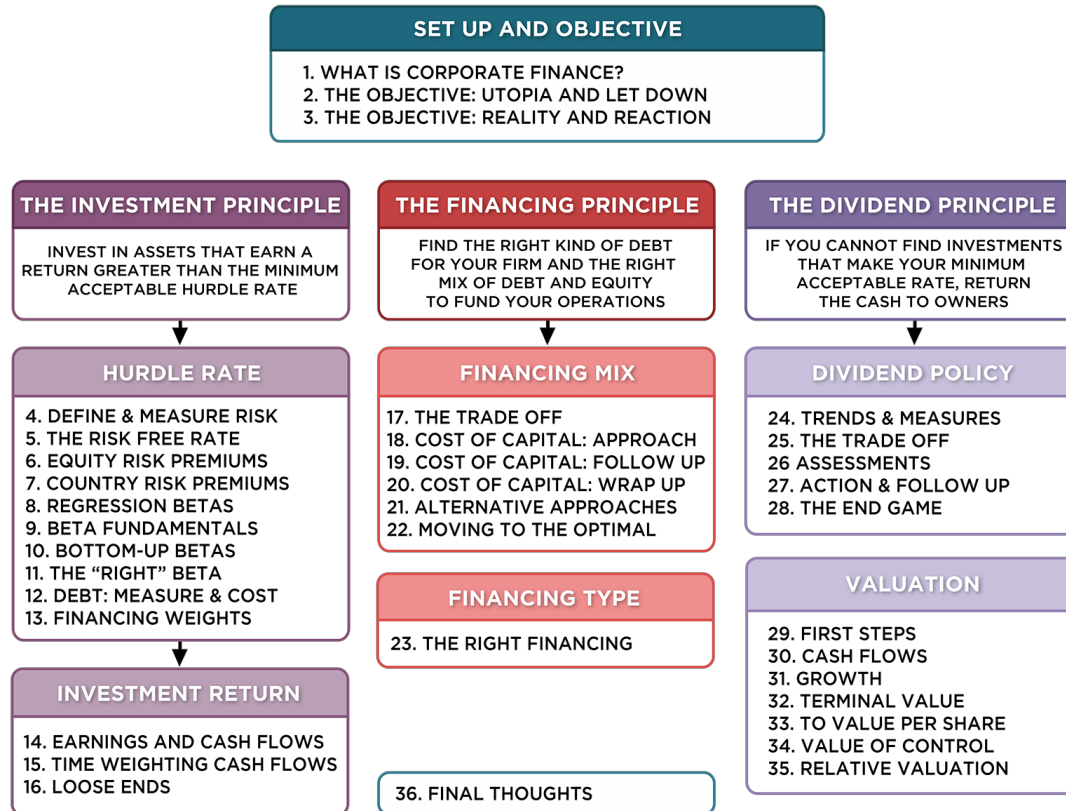


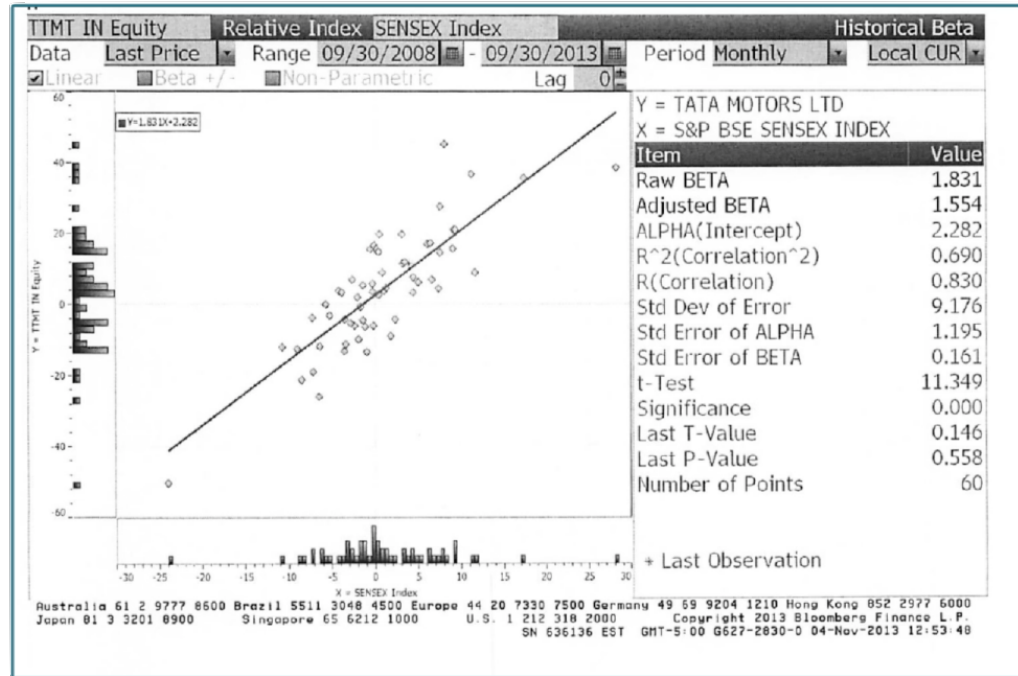
SESSION 9: BETA FUNDAMENTALS

Your business choices determine your risk profile!

SESSION 9: BETA FUNDAMENTALS



REGRESSION DIAGNOSTICS FOR TATA MOTORS



BETA = 1.83
67% RANGE
1.67-1.99

69% MARKET RISK
31% FIRM SPECIFIC

JENSEN'S α = $2.28\% - 4\%/12 (1 - 1.83) = 2.56\%$

ANNUALIZED = $(1 - 0.0256)^{12} - 1 = 35.42\%$

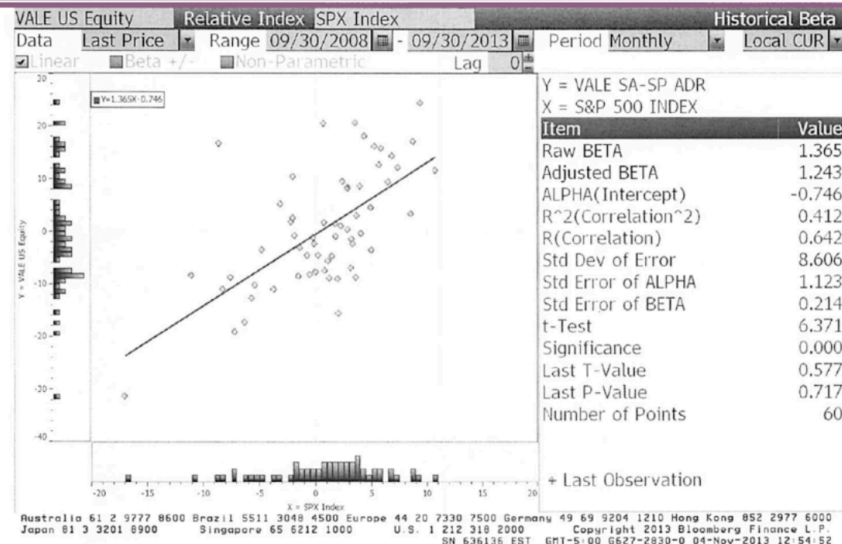
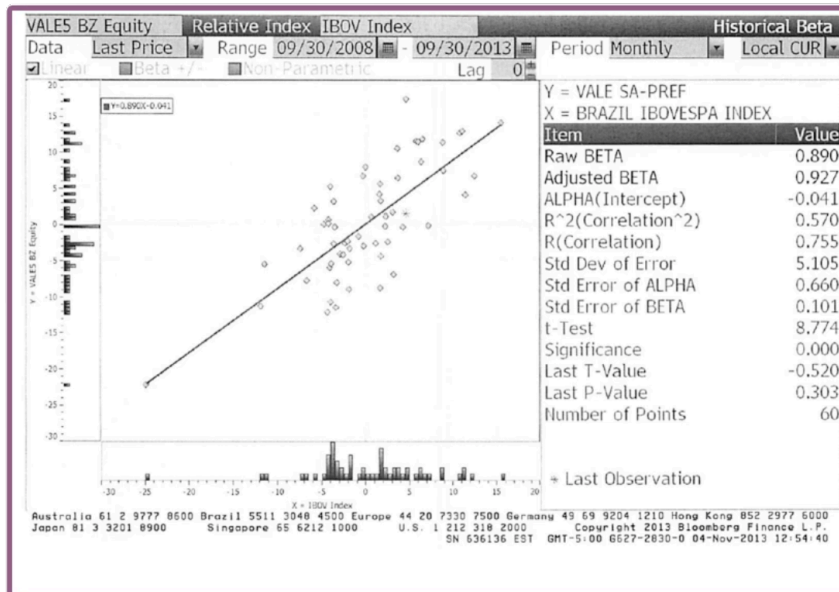
AVERAGE MONTHLY RISKFREE RATE (2008 - 13) = **4%**

EXPECTED RETURN (IN RUPEES)
= RISKFREE RATE + BETA * RISK PREMIUM

= $6.57\% + 1.83 (7.19\%) = 19.73\%$

BETA ESTIMATION AND INDEX CHOICE: VALE

REGRESSION DIAGNOSTICS FOR VALE



DEUTSCHE BANK AND BAIDU: INDEX EFFECTS ON RISK PARAMETERS

- For Deutsche Bank, a widely held European stock, we tried both the DAX (German index) and the FTSE European index.

	<i>DAX</i>	<i>FTSE Euro 100</i>
Intercept	-0.90%	-0.15%
Beta	1.58	1.98
Std Error of beta	0.21	0.29
R^2	51%	29%

- For Baidu, a NASDAQ listed stock, we ran regressions against both the S&P 500 and the NASDAQ.

	<i>S&P 500</i>	<i>NASDAQ</i>
Intercept	2.84%	2.15%
Beta	1.63	1.65
Std Error of beta	0.28	0.23
R^2	37%	47%

BETA: EXPLORING FUNDAMENTALS

BETA >2	BULGARI 2.45 
BETA BETWEEN 1 AND 2	QWEST COMMUNICATIONS 1.85 
	MICROSOFT 1.25 
	G.E. 1.25 
BETA < 1	EXXON MOBIL 0.70 
	ALTRIA (PHILIP MORRIS) 0.60 
BETA <0	HARMONY GOLD MINING -0.15 

DETERMINANT 1: PRODUCT TYPE

- Industry Effects: The beta value for a firm depends upon the sensitivity of the demand for its products and services and of its costs to macroeconomic factors that affect the overall market.
 - Cyclical companies have higher betas than non-cyclical firms
 - Firms which sell more discretionary products will have higher betas than firms that sell less discretionary products

DETERMINANT 2: OPERATING LEVERAGE EFFECTS

- Operating leverage refers to the proportion of the total costs of the firm that are fixed.
- Other things remaining equal, higher operating leverage results in greater earnings variability which in turn results in higher betas.

MEASURES OF OPERATING LEVERAGE

- Fixed Costs Measure = Fixed Costs / Variable Costs
 - This measures the relationship between fixed and variable costs. The higher the proportion, the higher the operating leverage.
- EBIT Variability Measure = % Change in EBIT / % Change in Revenues
 - This measures how quickly the earnings before interest and taxes changes as revenue changes. The higher this number, the greater the operating leverage.

DISNEY'S OPERATING LEVERAGE: 1987 - 2013

Year	Net Sales	% Change in Sales	EBIT	% Change in EBIT
1987	\$2,877		\$756	
1988	\$3,438	19.50%	\$848	12.17%
1989	\$4,594	33.62%	\$1,177	38.80%
1990	\$5,844	27.21%	\$1,368	16.23%
1991	\$6,182	5.78%	\$1,124	-17.84%
1992	\$7,504	21.38%	\$1,287	14.50%
1993	\$8,529	13.66%	\$1,560	21.21%
1994	\$10,055	17.89%	\$1,804	15.64%
1995	\$12,112	20.46%	\$2,262	25.39%
1996	\$18,739	54.71%	\$3,024	33.69%
1997	\$22,473	19.93%	\$3,945	30.46%
1998	\$22,976	2.24%	\$3,843	-2.59%
1999	\$23,435	2.00%	\$3,580	-6.84%
2000	\$25,418	8.46%	\$2,525	-29.47%
2001	\$25,172	-0.97%	\$2,832	12.16%
2002	\$25,329	0.62%	\$2,384	-15.82%
2003	\$27,061	6.84%	\$2,713	13.80%
2004	\$30,752	13.64%	\$4,048	49.21%
2005	\$31,944	3.88%	\$4,107	1.46%
2006	\$33,747	5.64%	\$5,355	30.39%
2007	\$35,510	5.22%	\$6,829	27.53%
2008	\$37,843	6.57%	\$7,404	8.42%
2009	\$36,149	-4.48%	\$5,697	-23.06%
2010	\$38,063	5.29%	\$6,726	18.06%
2011	\$40,893	7.44%	\$7,781	15.69%
2012	\$42,278	3.39%	\$8,863	13.91%
2013	\$45,041	6.54%	\$9,450	6.62%
Operating Leverage				
Average: 87-13		11.79%		11.91%
Average: 96-13		8.16%		10.20%
				11.91/11.79 = 1.01
				10.20/8.16 = 1.25

Average across entertainment companies = 1.35

Given Disney's operating leverage measures (1.01 or 1.25), would you expect Disney to have a higher or a lower beta than other entertainment companies?

- a) Higher
- b) Lower
- c) No Effect

DETERMINANT 3: FINANCIAL LEVERAGE

- As firms borrow, they create fixed costs (interest payments) that make their earnings to equity investors more volatile.
- This increased earnings volatility which increases the equity beta.
- The beta of equity alone can be written as a function of the unlevered beta and the debt-equity ratio.
- $\beta_L = \beta_u (1 + ((1-t)D/E))$

Where:

- β_L = Levered or Equity Beta
 - β_u = Unlevered or Asset Beta
 - D/E = Market value Debt to equity ratio
 - t = Marginal tax rate
- Earlier, we estimated the beta for Disney from a regression. Was that beta a levered or unlevered beta?
 - a. Levered
 - b. Unlevered

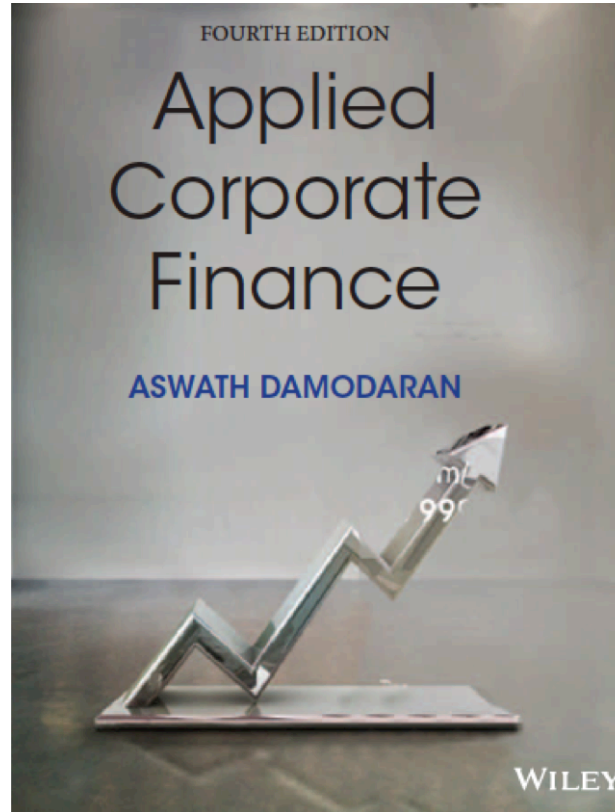
EFFECTS OF LEVERAGE ON BETAS: DISNEY

- The regression beta for Disney is 1.25. This beta is a levered beta (because it is based on stock prices, which reflect leverage) and the leverage implicit in the beta estimate is the average market debt equity ratio during the period of the regression (2008 to 2013)
- The average debt equity ratio during this period was 19.44%.
- The unlevered beta for Disney can then be estimated (using a marginal tax rate of 36.1%)
 - = Current Beta / (1 + (1 – tax rate) (Average Debt/Equity))
 - = $1.25 / (1 + (1 - 0.361)(0.1944)) = 1.1119$

DISNEY: BETA AND FINANCIAL LEVERAGE

<i>Debt to Capital</i>	<i>Debt/Equity Ratio</i>	<i>Beta</i>	<i>Effect of Leverage</i>
0.00%	0.00%	1.11	0.00
10.00%	11.11%	1.1908	0.08
20.00%	25.00%	1.29	0.18
30.00%	42.86%	1.42	0.30
40.00%	66.67%	1.59	0.47
50.00%	100.00%	1.82	0.71
60.00%	150.00%	2.18	1.07
70.00%	233.33%	2.77	1.66
80.00%	400.00%	3.95	2.84
90.00%	900.00%	7.51	6.39

Task
Evaluate the
business risk,
operating
leverage and
financial
leverage for
your company



Optional:
Read
Chapter 4